

Tooling List

Tool List No.	116A4141-203 OP40 REV03			Part name:	HINGE RIB				
Unit:	MM			Part Number:	116A4141-203				
									
Tool No.	Tool Name	Tool Description	Insert	Tool Diameter (D1)	Flute Length (L1)	Tool Ext. Length (L2)	Arbor Description	Tool Path Time in Minutes	Remarks/ Tool Teach Length =
T30	Endmill Ø20 R0.8 x 3-Flute (Solid Rougher Tool) (PGTC)	00001048 (ROUGH-REE D20 X CL48 X OAL100 X SHK20 X 3 FLUTE)	N/A	20	48	76	A63.140.20.2 Ø20 x 100 (Haimer) Through Coolant Arbor	1.00	176 ± 1
T31	Endmill Ø16 R0.8 x 3-Flute (Solid Rougher Tool) (PGTC)	00001049 (ROUGH-REE D16 X CL42 X OAL100 X SHK16 X3FL)	N/A	16	42	63	A63.140.16.2 Ø16 x 95 (Haimer) Through Coolant Arbor	15.00	158 ± 1
T32	Endmill Ø12 R0.8 x 3-Flute (Solid Rougher Tool) (PGTC)	00001050 (ROUGH-REE D12 X CL30 X OAL80 X SHK12 X3FL)	N/A	12	30	55	A63.140.12.2 Ø12 x 90 (Haimer) Through Coolant Arbor	6.00	145 ± 1
T36	Endmill Ø20 R0 x 3-Flute (PGTC)	00001173 (DLC-D20 X CL100 X OAL160 X SHK20 X 3 FLUTE)	N/A	20	100	117	HSK63AHCTHT20090M Ø20 x 90 (Kennametal)	5.00	207 ± 1
T37	Endmill Ø16 R1.0 x 3-Flute (Solid Rougher Tool) (PGTC)	00000804 (DLC-D16 R1 X FL50 X OAL 110 X SHK DIA 16)	N/A	16	50	86	A63.144.16.2 Ø16 x 130 (Haimer) Through Coolant Arbor	6.00	196 ± 1
T38	Endmill Ø20 R1.0 Minimaster x 3-Flute (Iscar-PGTC)	Holder Ø18.3 x Ø20 x NL80: 3104447 MM S-A-L130-C20-T12-W-H (Through-Coolant)	Insert Ø20 R1.0: 5622017 MM EA200B12R1.0-3T12	20	12.7	123	HSK63AHCTHT20090M Ø20 x 90 (Kennametal)	3.00	213 ± 1
T40	Endmill Ø16 R3.0 x 3-Flute (PGTC)	00001172 (DLC-D16 X R3 X CL50 X OAL110 X 3 FLUTE)	N/A	16	50	72	A63.140.16.00 Ø16 x 95 (Haimer)	35.00	167 ± 1
T41	Endmill Ø12 R3.0 (PGTC)	00001000 (DLC-EDS-D12 X R3X CL45X OAL110 X FL3)	N/A	12	45	62	A63.144.12.2 Ø12 x 130 (Haimer) Through Coolant Arbor	12.00	192 ± 1
T43	Drill Ø6.361 (PGTC)	00000772 (BURNISHING DRILL D6.361 +/- 0.002 X CL25 X OAL70X SHK8)	N/A	6.361	25	50	A63.144.08 Ø8 x 130 (Haimer)	0.00	180 ± 1
T45	Endmill Ø12 R3.0 x 2-Flute (PGTC)	00001175 (DLC-EDS-D12 X R3X CL14 X NL40 X OAL90XFL2)	N/A	12	14	60	A63.144.12.2 Ø12 x 130 (Haimer) Through Coolant Arbor	21.00	190 ± 1
T48	Drill Ø3.26 x 2-Flute (PGTC)	00000796 (D3.26 +/- 0.005 X CL20 X OAL60 XSHK6)	N/A	3.26	20	60	A63.140.06.00 Ø6 x 80 (Haimer)	1.00	140 ± 1
T49	Drill Ø2.45 x 2-Flute (PGTC)	00000827 (CARBIDE DRILL-D2.45 X CL25 X OAL70 X SHK 6)	N/A	2.45	25	55	A63.144.08 Ø6 x 130 (Haimer)	1.00	185 ± 1
T50	Drill Ø9.51 (PGTC)	00001160 (BURNISHING DRILL D9.512 + 0.005 , - 0.00 X CL35 X NL5 X OAL90 X SHK10)	N/A	9.51	30	60	A63.140.10.00 Ø10 x 85 (Haimer)	0.00	145 ± 1
T51	Reamer Ø17.463 x 6-Flute (PGTC)	00001145 (BURNISHING DRILL D17.463 + 0.010 , - 0.00 X CL30 X OAL90 X SHK20)	N/A	17.463	30	55	A63.144.20 Ø20 x 130 (Haimer)	0.00	185 ± 1
T52	Drill Ø7.92 (PGTC)	00001159 (BURNISHING DRILL D7.920 + 0.005 , - 0.00 X CL35 X OAL90 X SHK8)	N/A	7.92	30	50	A63.144.08 Ø8 x 130 (Haimer)	0.00	180 ± 1
T53	Ballnose Ø8 R4.0 x 2-Flute (PGTC)	00001176 (BALLNOSE-D8 X R4 X OAL80 X FL14 X SHK8 X 2FL)	N/A	8	14	50	A63.144.08 Ø8 x 130 (Haimer)	5.00	180 ± 1
T54	Endmill Ø12 R0.0 x 3-Flute (PGTC)	00000840 (DLC-D12 X FL50 X OAL 110 X SHK12 X 3FL)	N/A	12	50	65	HSK63AHC12090M Ø12 x 90 (Kennametal)	5.00	155 ± 1
T58	Endmill Ø20 R0.5 Minimaster x 3-Flute (PGTC)	Holder Ø18.3 x Ø20: 3104448 MM S-A-L200-C20-T12-W-H (Through-Coolant)	5668964 MM EA200H20R0.5CF-3T12	20	20	183	HSK63AHCTHT20090M Ø20 x 90 (Kennametal)	5.00	273 ± 1
T61	Endmill Ø20 R1.0 x 3-Flute (PGTC)	00001008 (D20 X FL180 X OAL250 X SHK20 X 3FL ENDMILL)	N/A	20	180	198	HSK63AHCTHT20090M Ø20 x 90 (Kennametal)	5.00	288 ± 1